| | | |
|---|---|---|
| | | $P \quad 3.0 \times 10^5$ |
| **21** | **C** | For an ideal gas, $pV \propto T$<br>So, $\frac{T_f}{T_i} = \frac{p_f}{p_i}\frac{V_f}{V_i} = 2 \times 2 = 4$.<br>Also, the average KE of a molecule of an ideal gas = $\frac{1}{2}mv_{rms}^2 = \frac{3}{2}kT$<br>So, $v_{rms}^2 \propto T$<br>$\left(\frac{v_f}{v_i}\right)^2 = \frac{T_f}{T_i} = 4$<br>$v_f = 2v_i = 2u$ |
| **22** | **C** | |